OUR PUBLICATION OPTIONS

Choose the study we can support

We have three publication paths: evaluate the proposed framework, test operational irrigation outcomes, or study how AI relates to professional decision-making.

1	2	3
Proposed framework	**Operational study**	**Governance study**
Design and evaluate the idea	Use existing data or a pilot	Study professional perceptions

1 An evaluated proposed framework

Question: Can our design turn complete or incomplete site records into a clear, traceable irrigation decision? We compare its requirements and features with related designs, test defined scenarios, and use expert feedback to improve it. The result is an evaluated design, not a measured water-saving effect. [7]

2 An operational evaluation

Question: Does our irrigation process reduce water supplied per square metre while meeting a predefined plant-condition criterion? We can analyse an existing deployment if its records are usable, or collect the measurements through a focused pilot. The comparison and study design determine how confidently we can attribute a change to the system.

3 A governance study

Question: How is perceived AI capability associated with management practice and decision quality among the professionals we study? We use verified survey responses and relevant interviews to examine these relationships and adoption needs. This route requires suitable measures and original participant records. [11]

What we keep from the current paper

We retain the Green Riyadh context, the connection between sensors and executive decisions, and any traceable data or interviews. We focus the first study on irrigation. Cooling and air quality can remain future extensions unless we have suitable measurements to evaluate them.

The following sections assess the current evidence and develop the framework and operational designs. The research plan then sets out the governance option and the evidence needed to choose our path.

PUBLISHED EXAMPLES

What comparable studies actually did

These papers give us concrete models for the research contribution, the scale of a pilot and the way measurements connect to decisions.

Two models for a proposed framework

Rooting for sustainable smart cities (Electronic Markets, 2026) develops irrigation decision support through practitioner requirements, design features and an evaluated prototype. It is a close model for turning our topic into a specific design contribution. [2]

GreenTwin (IBDAP, 2026) presents a digital-twin reference architecture and evaluates it through requirements traceability and comparison with existing platforms. We can adopt this pattern: state the requirements, show how our design meets them, and test its distinctive features. [3]

Three useful models for collecting real data

Published study	Scale and measurements	Lesson for our study
Gimpel et al. [4] *Designing smart and sustainable irrigation*	18 sensors at 8 urban trees; a three-month prototype; six interviews involving five experts.	A small, well-documented study can support a design contribution. Combine physical measurements with requirements from practitioners.
Gasser et al. [5] *Water efficiency in smart cities*	24 parcels, but only 3 soil-monitoring locations; hourly soil data and main-valve flow every second. Recommended irrigation was compared with the original schedule.	Useful for inputs and modelling. The 24 parcels are not 24 separately instrumented replicates; this was not a randomised treatment trial.
Nautiyal et al. [6] *Evaluation of Smart Irrigation Technologies*	24 residential sites in six geographical clusters, with four treatments. Weekly meter readings, visual turf quality and NDVI.	Use contemporaneous comparison sites and measure plant condition alongside water consumption.

The literature supports these design principles, not a universal sample count. Our 24-zone, 12-week plan below is a feasibility proposal; its final size and duration depend on local variability and the precision required. Repeated readings do not create extra treatment replicates. [8]

ASSESSMENT OF OUR MANUSCRIPT

The evidence needs a clearer story

Our main issue is that the manuscript describes both a proposed design and achieved improvements. We need to choose one account and support it consistently.

> *"The framework significantly attained improvement of 20-30% in water-use efficiency and increased speed of executive planning by a maximum of 25%."*

Manuscript excerpt - Abstract [1]

This requires measured water volumes, a defined comparison, and timed planning tasks. The manuscript currently shows ten sensor rows, five district temperature summaries and interview themes, without the underlying analysis needed for those percentages. [1]

> *"This frame is ready for NVivo coding and quote-mapping:"*

Manuscript excerpt - Results, Qualitative results [1]

The methods also say that interview analysis was completed. We need to establish the actual status of the ten interviews and use the records to support whichever account is correct.

What the table records

24.2% lower flow at the later observation

The table does not report how long irrigation ran.

The flow difference is a useful calculation, but water use also depends on runtime. For example, **125.7 L/min for 30 minutes = 3,771 L**, while **95.3 L/min for 45 minutes = 4,288.5 L**. Those durations are illustrative assumptions: they show why we must collect total volume or start/end times.

The temperature table lacks a measurement period and comparison design. The 25% green-cover and 22% PM2.5 claims also need their source data and calculations. We should retain these outcomes only when that evidence is available.

PROPOSED FRAMEWORK PATH

Use our current data to test the design

Our design question is whether the framework produces an understandable, justified decision when data are sufficient, incomplete or conflicting. We evaluate that behaviour before making any claim about field performance. [7]

What we build

A decision-support prototype that checks records, applies specified irrigation rules and records human approval. Define a moisture target from soil and species needs, a water-budget constraint, and a rule for missing inputs. Include a comparison of distinctive features with related work. Describe AI or a digital twin as implemented only if its model and evaluation actually exist.

What data we use and how much

Input	Our working plan	Purpose
Existing sensor records	Use all 10 rows after documenting their origin and units.	Test import, date parsing, units and source tracking. These rows cannot train or validate an irrigation forecast.
Existing district table	Use the 5 rows as a display example once their origin is established.	Show how the interface reports missing dates or context instead of presenting an effect estimate.
Scenario cases	Start with 3 scenario families: complete inputs; missing/invalid values; conflicting moisture and water-budget signals. Test relevant boundary variants.	Specify expected action and pass/fail rule first; test missing volume, invalid units, stale readings and manual override.
Expert feedback	Start with 6-8 experts across irrigation, planning and data systems. Review role coverage and information gained; extend where gaps remain. [9]	Assess completeness, usability and missing controls. Reuse the existing interviews only if their records and questions fit this purpose.

Save **requirement -> component -> test -> expected action -> actual result -> revision**. For example, missing volume must block a water-saving calculation; an unreliable moisture reading must trigger a data check. Label added values synthetic. Code expert comments into requirements, issues and revisions. [7,9]

Our result: demonstrated workflow behaviour and design improvements. Expert approval alone does not establish water savings. Novelty comes from the design gap addressed, supported by the comparison with related work.

PILOT DESIGN

How much data should we collect?

Our field question is whether the proposed irrigation process lowers water supplied per square metre while meeting a plant-condition criterion. Each zone needs separate control and a meter so we can attribute its water use.

24
planned zone units

84
days per zone

2,016
daily analysis rows

Our starting design

Budget for **24 zones in 12 matched pairs**. Match area, species/age, soil and exposure. Observe current practice for **4 weeks**; then use the proposed sensor-guided process in one zone per pair for **8 weeks**, with current practice in the other. Randomise within pairs where feasible. Keep other care comparable and log recommendations, actual watering and overrides.

These counts and durations are **our planning choices**. Before the comparison, use pilot/prior variability, pairing, missing-data allowance and a meaningful target effect to calculate power or expected precision. If resources allow only this scale, frame it as a feasibility pilot with an effect estimate and interval. [8,10]

Collection schedule

Measurement	Frequency and planned count	Why we need it
Water delivered	Meter every event; aggregate daily. 24 x 84 = **2,016 scheduled zone-days**. Record zero only with valid evidence of no watering.	Our primary outcome: actual water use.
Soil moisture	Hourly at a representative root-zone position in each homogeneous zone. One probe per zone gives **48,384 scheduled readings** per parameter; placement/depth follows soil and roots.	Check plant water availability and support scheduling; add probes where soil or exposure varies.
Weather	Hourly from a representative local source; summarise daily.	Explain differences in demand caused by heat, rainfall and atmospheric conditions.
Plant condition	Select 5 tagged plants per zone; inspect weekly. **120 plants and 1,440 scheduled visits**. Increase coverage for heterogeneous planting.	Check that lower water use is accompanied by acceptable plant condition.

Analysis unit: the zone, with 12 paired comparisons, not 48,384 independent samples. If zones share control or treatment allocation, use that larger unit. Before intervention, define the stress score and acceptable increase in the fraction of stressed plants; a nonsignificant difference alone does not establish equivalence. [8,10]

DATA FILES

What the spreadsheets should look like

Keep one row for one defined observation. Use the same zone identifier across files so we can join water use, weather, plant condition and treatment status.

1 Zone register: one row per zone

zone_id	pair	group	area_m2	trees
Z01	P01	Intervention	200	20
Z02	P01	Comparison	210	21

Also record coordinates, species, planting age, soil, exposure, irrigation equipment, sensor location/depth and the exact intervention date. These explain how zones were matched and how water use should be normalised.

2 Daily analysis file: one row per zone per day

day	zone	phase	water_m3	rain_mm	mean_vwc_pct
D01	Z01	Baseline	1.20	0.0	19.0
D01	Z02	Baseline	1.25	0.0	18.8
D29	Z01	Intervention	0.95	0.0	19.2

These rows are synthetic examples. Also store actual date/time zone, coverage flags, maintenance/leaks and the local weather-source ID. Link rainfall and reference evapotranspiration from a separate dated weather file. Group is the assigned treatment; phase is baseline versus post-start. Use both in analysis.

3 Raw sensor and plant records: preserve the detail

File	One row represents	Key fields
sensor_log	One sensor at one timestamp	timestamp, zone_id, sensor_id, parameter, value, unit, quality_flag
irrigation_events	One watering event	event_id, zone_id, start/end, metered_volume and unit, rule_version, recommendation, actual_action, override
plant_health	One tagged plant at one visit	visit_date, zone_id, plant_id, alive, stress_score, assessor

Record moisture type: volumetric content (%) and water potential (kPa) are different. Define calibrated soil/species thresholds and the plant-stress scale before collection. Preserve raw exports and calibration. Aggregate sensors/events to zone-day before joining files; otherwise repeated rows can multiply water totals. A missing meter record remains missing, not zero.

ANALYSIS PLAN

From our files to a defensible answer

We use one primary analysis for the field pilot: compare area-normalised water use between paired zones over the same baseline and intervention periods. We evaluate plant condition alongside that result.

1 Check and assemble the dataset

Validate zone IDs, units, meter coverage and intervention dates; flag implausible readings, leaks and downtime. Build one row per zone-day by aggregating raw events and sensors, then join the register and weather data. Report completeness by group and period. Agree exclusions before viewing the treatment comparison.

2 Define the water outcome

Use **litres per square metre per day**: daily metered litres divided by zone area. Show volume totals as well. Summarise baseline and post-start means for each zone. Area normalisation complements matching; it does not by itself remove differences in species, age or exposure.

3 Compare changes within matched pairs

For each pair, calculate **(post minus baseline in the intervention zone) minus (post minus baseline in its comparison zone)**. Average the 12 pair differences with a 95% confidence interval that treats pairs as the replicates; inspect distributions and use a suitable paired or randomisation method. Negative values indicate a relative reduction. Pre-specify any weather adjustment. [8,10]

4 Check the interpretation

Examine baseline patterns, missingness, leaks and overrides. Show the control trend: water use fell historically in the control group too in Nautiyal et al., making an uncontrolled reduction hard to interpret. For nonrandom assignment, assess comparable trends and remaining confounding; describe an association where attribution is weak. [6]

5 Evaluate plant condition and report the conclusion

Compare zone-level stress summaries, survival and valid hours below the agronomic moisture threshold. Compare the upper confidence bound for added stress with the agreed deterioration limit. If uncertainty crosses that limit, report plant condition as inconclusive rather than maintained. [10]

Report: the absolute water difference and its interval, a clearly defined denominator for any percentage, plant-condition findings, data completeness and actual treatment adherence. The result applies to the tested process, zones and season. Existing deployment records follow this same sequence, with the analysis limited by the available comparison.

Select the path our evidence supports

The framework, operational evaluation and governance study answer different questions. We should choose a primary contribution and build the evidence needed for that contribution.

Working path	Evidence we need	What we can report
Proposed framework	Related-work comparison, explicit decision rules, worked scenarios and relevant expert feedback.	A defined design contribution, demonstrated behaviour and improvements after evaluation.
Operational evaluation	A documented deployment with usable records, or the controlled irrigation pilot described earlier.	Water-use differences, plant-condition findings and implementation lessons.
Governance study	Verified survey responses, the full questionnaire and scoring rules, recruitment information and suitable interview records.	Associations between perceived AI capability, management practice and decision quality.

How governance evidence can contribute

A complementary question is: **How is perceived AI capability associated with strategic decision quality among the professionals we study?** The existing strategic-management draft provides a starting point for this work. Its themes can also inform framework requirements for explainability, human oversight, training and accountability. [11,12]

For a survey study, define the population from verified recruitment records, confirm what each item measures, size the sample for the intended association model, then evaluate the scales before analysing associations. Interview evidence can explain adoption barriers and practical needs. These findings concern professional perceptions; environmental benefits require the operational measurements described earlier. [11]

The critical issue to resolve first

The current survey file repeats the same 30-case block ten times and contains identical questionnaire columns. We need the original response export and questionnaire before using it as empirical evidence. We should not treat its 300 rows as 300 independent respondents or assume that removing repetitions recovers a valid sample. [12]

Our next step

Bring together the original operational exports, questionnaire and response records, relevant interview material, and documentation of the implemented system. If usable operational data exist, proceed with the field analysis. If verified survey data exist, develop the governance study. Otherwise, prioritise the evaluated framework and its scenarios, with a pilot planned as the next stage.

READING LIST

Studies behind the plan

The references below support the examples, sampling precedents and analysis choices used in this report.

[1] Leveraging AIoT for Data-Driven Executive Decision Making in Sustainable Urban Development. Green Riyadh research manuscript. Internal study material.

[2] Rooting for sustainable smart cities: Designing smart irrigation for urban green spaces. Rebholz, D., Schnaak, F., & Gimpel, H. (2026). Electronic Markets, 36, 72. DOI: 10.1007/s12525-026-00926-8.

[3] GreenTwin: a Cloud-Native Digital Twin Architecture for Citizen-Centric Green Interventions in Smart Cities. IBDAP, 2026. Architecture evaluation through traceability and comparison. DOI: 10.1109/IBDAP69915.2026.11677757.

[4] Designing smart and sustainable irrigation: A case study. Gimpel, H., Graf-Drasch, V., Hawlitschek, F., & Neumeier, K. (2021). Journal of Cleaner Production, 315, 128048. DOI: 10.1016/j.jclepro.2021.128048.

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[11] Best Practices for Developing and Validating Scales for Health, Social, and Behavioral Research: A Primer. Boateng et al. (2018). Frontiers in Public Health, 6, 149. Scale development, reliability and validity. DOI: 10.3389/fpubh.2018.00149.

[12] AI in Strategic Management to Support Saudi Vision 2030 Transformation. Research draft and accompanying AI-SM-DM survey data. Internal study materials.